What Replaces the Inner Product over a Finite Field

and what the opposite category of linear algebra is

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Abstract

Part I asks what survives of the inner product when the ground field is finite. Separating *nondegeneracy* from *positive-definiteness*, we find the first survives in abundance and the second cannot even be stated; what is lost is the orthogonal splitting $V = W \perp W^{\perp}$, and what is gained is isotropy, hence self-dual codes, polar geometries and the finite classical groups.

Part II asks a categorical question with the same shape. The opposite of the category of sets is the category of complete atomic Boolean algebras. What, then, is the opposite of the category of vector spaces over $\mathbb{F}_q$? The answer forks: as a bare category $\mathbf{FinVect}_q$ is self-dual, but if one deforms the *recipe* behind $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$ — take subobject lattices — one lands on projective geometry, whose anti-automorphisms are classified by exactly the three families of forms found in Part I. The two halves of the document turn out to be one theorem seen twice — and, by Piron's and Solèr's reconstruction theorems, a third time, as the list of possible quantum mechanics, the symplectic case being excluded by precisely the definiteness Part I is about losing.

Part III takes $q \to 1$. Isotropy vanishes, the polarity $W \mapsto W^{\perp}$ becomes Boolean complementation, projective geometry becomes the Boolean lattice, and we recover $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$. We also separate the field with one element from the prime at infinity, which are commonly and productively confused.

Everything is defined from scratch. A reader who knows what a set is can read this straight through.

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Part I
Forms over a finite field

1 The mishnah: what "naturally isomorphic" is doing here

1.1 The starting objects

Definition 1.1 (Field)**.** A *field* k is a set with two operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributing over $+$, with additive identity 0 and multiplicative identity $1 \neq 0$, such that every element has an additive inverse and every nonzero element has a multiplicative inverse. Examples: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and for each prime p the set $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ of integers modulo p.

Definition 1.2 (Finite field)**.** For each prime power $q = p^e$ there is, up to isomorphism, exactly one field with q elements, written $\mathbb{F}_q$. Its *characteristic* is p, meaning $\underbrace{1 + \cdots + 1}_{p} = 0$. Its multiplicative group $\mathbb{F}_q^\times = \mathbb{F}_q \setminus \{0\}$ is cyclic of order $q-1$. No field has a number of elements that is not a prime power.

Definition 1.3 (Vector space, linear map, dual)**.** A *vector space* over k is an abelian group V with a scalar multiplication $k \times V \to V$ satisfying the usual axioms. A *linear map* $f : V \to W$ satisfies $f(u+v) = f(u) + f(v)$ and $f(\lambda v) = \lambda f(v)$. The *dual space* is

$$V^* \;=\; \mathrm{Hom}_k(V, k) \;=\; \{\text{linear maps } V \to k\},$$

itself a vector space. If $\dim V = n < \infty$ then $\dim V^* = n$, so $V \cong V^*$ — but only after choosing a basis.

Definition 1.4 (Bilinear form)**.** A *bilinear form* on a k-vector space V is a map $\langle -, - \rangle : V \times V \to k$ linear in each argument separately. It is *symmetric* if $\langle v, w\rangle = \langle w, v\rangle$, *alternating* if $\langle v, v\rangle = 0$ for all v, and *nondegenerate* if $\langle v, w\rangle = 0$ for all w forces $v = 0$.

Definition 1.5 (Gram matrix)**.** Given a basis $e_1, \dots, e_n$ of V, the *Gram matrix* is $G = (\langle e_i, e_j\rangle)_{i,j}$. The form is nondegenerate iff $\det G \neq 0$. Changing basis by an invertible P replaces G by $P^{\mathsf{T}} G P$, so $\det G$ changes by a nonzero square.

1.2 The two ingredients of an inner product

Let V be a finite-dimensional complex inner product space. The form gives a map

$$\flat : V \longrightarrow V^*, \qquad v \longmapsto \langle -, v\rangle,$$

conjugate-linear (or linear into the conjugate dual $\overline{V}^*$, depending on convention), and it is an isomorphism because the form is *nondegenerate*. Positive-definiteness — the requirement $\langle v, v\rangle > 0$ for $v \neq 0$ — is sufficient for nondegeneracy but by no means necessary. So the isomorphism $V \cong V^*$ consumes only a fraction of what an inner product supplies.

Two remarks on the word "natural", both of which sharpen the question. (Naturality is defined precisely in §8; the reader may take the following on trust for now and return.)

Remark 1.6. There is no natural transformation $\mathrm{Id} \Rightarrow (-)^*$ on finite-dimensional vector spaces: $(-)^*$ is contravariant, so the statement does not typecheck as naturality. What one has is *canonicity relative to a chosen form*, which has exactly the same epistemic status as a chosen basis. The genuinely natural isomorphism is $V \cong V^{**}$.

Remark 1.7. Consequently the right question is not "can we define an inner product over $\mathbb{F}_q$?" but: *which of the two ingredients survives?* Nondegeneracy is pure linear algebra and survives. Definiteness is a statement about an *ordering* of the ground field, and $\mathbb{F}_q$ admits none compatible with its arithmetic: in an ordered field $1 > 0$ forces $1 + 1 + \cdots > 0$, whereas in $\mathbb{F}_p$ the sum of p copies of 1 is 0.

2 What can be defined over $\mathbb{F}_q$

Three families of nondegenerate forms exist, and — unlike over $\mathbb{R}$ or $\mathbb{C}$ — all three are classified completely. Finite fields are, in this respect, *nicer* than the classical ground fields.

2.1 Symmetric bilinear and quadratic forms

Definition 2.1 (Isometry)**.** Two forms $(V, \langle -,-\rangle_V)$ and $(W, \langle -,-\rangle_W)$ are *isometric* if there is a linear isomorphism $f : V \to W$ with $\langle f(u), f(v)\rangle_W = \langle u, v\rangle_V$. The *isometry group* of a form is the group of self-isometries.

Definition 2.2 (Discriminant)**.** For a nondegenerate symmetric form with Gram matrix G, the *discriminant* is the class of $\det G$ in $k^\times/(k^\times)^2$. By the change-of-basis rule above this is independent of the basis.

For q odd, $\mathbb{F}_q^\times$ is cyclic of even order $q-1$, so squares have index 2 and $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^2 \cong \mathbb{Z}/2$.

Theorem 2.3 (Classification, q odd)**.** *Nondegenerate symmetric bilinear forms on $\mathbb{F}_q^n$ are classified up to isometry by the discriminant, an element of $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^2 \cong \mathbb{Z}/2$. There are exactly two isometry classes in each dimension.*

Compare: over $\mathbb{C}$ there is exactly one class in each dimension (Gram matrix I); over $\mathbb{R}$, Sylvester's law of inertia gives $n+1$ classes indexed by signature. Finite fields sit strictly between, retaining a vestigial ghost of signature of size 2. The isometry groups are the orthogonal groups $O_n^+(q)$ and $O_n^-(q)$.

Remark 2.4 (Characteristic two). In characteristic 2 the symmetric and alternating conditions collide, since $x = -x$. The correct object is then the *quadratic form* $Q : V \to k$ (satisfying $Q(\lambda v) = \lambda^2 Q(v)$ and with $B(x,y) = Q(x+y) - Q(x) - Q(y)$ bilinear) rather than its *polarisation* B, which is alternating and carries strictly less information. Read everything below about symmetric forms with q odd unless stated otherwise.

2.2 Alternating (symplectic) forms

Nondegenerate alternating forms exist only in even dimension, and there is exactly *one* up to isometry: an orthogonal sum of hyperbolic planes (Definition 4.1). The isometry group is the symplectic group $\mathrm{Sp}_{2n}(q)$. The classification is even coarser than over $\mathbb{C}$: symplectic geometry has no local invariants at all.

2.3 Hermitian forms: the honest analogue

This is the closest finite-field relative of the complex inner product.

Definition 2.5 (Galois group, Frobenius)**.** For a field extension $L \supseteq k$, the *Galois group* $\mathrm{Gal}(L/k)$ is the group of field automorphisms of L fixing k pointwise. For finite fields, $\mathrm{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q)$ is cyclic of order d, generated by the *Frobenius* $\sigma : x \mapsto x^q$.

Complex conjugation is the nontrivial element of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, a group of order two. The finite analogue is immediate: take $\mathbb{F}_{q^2} \supset \mathbb{F}_q$, whose Galois group is generated by $\sigma : x \mapsto x^q$, an involution ($\sigma^2 = \mathrm{id}$ since $x^{q^2} = x$ on $\mathbb{F}_{q^2}$) with fixed field exactly $\mathbb{F}_q$ (the roots of $x^q = x$).

Definition 2.6 (Semilinear, sesquilinear, Hermitian)**.** Let σ be a field automorphism of L. A map f is *σ-semilinear* if $f(u+v) = f(u) + f(v)$ and $f(\lambda v) = \sigma(\lambda) f(v)$. A *$\sigma$-sesquilinear form* is $\langle -, - \rangle : V \times V \to L$ linear in the first argument and σ-semilinear in the second. It is *Hermitian* if σ is an involution and $\langle w, v \rangle = \sigma \langle v, w \rangle$.

Proposition 2.7. *For a Hermitian form over $\mathbb{F}_{q^2}$ and any $v \in V$ we have $\langle v, v \rangle^q = \langle v, v \rangle$, hence $\langle v, v \rangle \in \mathbb{F}_q$.*

That is: *the norm is always real*, exactly as over $\mathbb{C}$. Moreover there is exactly one nondegenerate Hermitian form per dimension up to isometry, with Gram matrix I; the isometry group is the unitary group $U_n(q)$. Every structural feature of the complex inner product is reproduced save one: $\mathbb{F}_q$ is not ordered, so "$\langle v, v \rangle > 0$" is meaningless.

Together with $\mathrm{GL}_n(q)$, these isometry groups are the *finite classical groups*. The question "what replaces the inner product over $\mathbb{F}_q$?" is therefore an entrance to the classification of finite simple groups.

3 The objection: what actually breaks

Nondegeneracy buys $V \cong V^*$. It does *not* buy the orthogonal splitting.

Definition 3.1 (Orthogonal complement, isotropy)**.** For a subspace $W \subseteq V$ put $W^\perp = \{v \in V : \langle v, w \rangle = 0 \ \forall w \in W\}$. A vector $v \neq 0$ is *isotropic* if $\langle v, v \rangle = 0$; a subspace is *totally isotropic* if the form vanishes identically on it; a form is *anisotropic* if it has no isotropic vectors.

If v is isotropic then $v \in \langle v \rangle \cap \langle v \rangle^\perp$, so $V \neq \langle v \rangle \oplus \langle v \rangle^\perp$. Gram–Schmidt fails: one cannot normalise a vector of norm zero, nor divide by it. Orthogonal projection, least squares and the analytic spectral theorem fail with it. There is no norm, no metric, no completeness, no analysis.

Worse — or better — isotropy is unavoidable.

Theorem 3.2 (Chevalley–Warning, quadratic case)**.** *Every quadratic form over $\mathbb{F}_q$ in $n \geq 3$ variables has a nontrivial zero. Consequently an anisotropic quadratic space over $\mathbb{F}_q$ has dimension at most 2.*

Over $\mathbb{R}$, anisotropic spaces exist in every dimension — that is precisely what positive-definiteness means. Over $\mathbb{F}_q$ they are confined to dimension ≤ 2.

What does survive, from nondegeneracy alone, is the numerical duality:

$$\dim W + \dim W^\perp = \dim V, \qquad (W^\perp)^\perp = W.$$

One keeps the duality and loses the splitting. This sentence is the whole of Part I in compressed form, and Part II is its categorical restatement.

4 Attempted resolution: Witt's theory

Definition 4.1 (Hyperbolic plane)**.** The *hyperbolic plane* $\mathbb{H}$ has basis e, f with $\langle e, e \rangle = \langle f, f \rangle = 0$ and $\langle e, f \rangle = 1$. It is nondegenerate and contains isotropic vectors by construction.

Theorem 4.2 (Witt decomposition)**.** *Every nondegenerate space decomposes as*

$$V \cong \underbrace{\mathbb{H} \perp \cdots \perp \mathbb{H}}_{m} \perp V_{\mathrm{an}},$$

with V_{an} anisotropic and unique up to isometry. The integer m is the Witt index, the common dimension of the maximal totally isotropic subspaces.

Theorem 4.3 (Witt extension)**.** *Any isometry between subspaces of a nondegenerate space extends to an isometry of the whole space.*

Witt extension replaces the statement that the unitary group acts transitively on orthonormal frames: the isometry group acts transitively on isometric configurations. Over $\mathbb{F}_q$, since $\dim V_{\mathrm{an}} \leq 2$, finite quadratic spaces are *almost entirely hyperbolic*. A situation that from the $\mathbb{C}$ side looks maximally degenerate is in fact the source of all the interesting combinatorics.

5 *Mai nafka minah*: what changes in practice

5.1 Coding theory and self-duality

Definition 5.1 (Linear code)**.** A *linear code* of length n over $\mathbb{F}_q$ is a subspace $C \subseteq \mathbb{F}_q^n$. Its *dual* is $C^\perp$ taken with respect to the standard bilinear form $\sum_i x_i y_i$.

Self-dual codes, $C = C^\perp$, exist — the extended binary Golay code $[24, 12, 8]$, the hexacode over $\mathbb{F}_4$ — and they are the most beautiful objects in the subject. Over $\mathbb{R}$ with a positive-definite form this is flatly impossible: $W = W^\perp$ forces $W = 0$. Self-duality is a gift of isotropy.

Warning 5.2. Self-duality is not special to finite fields; it is special to *indefinite or unordered* ground fields. Over $\mathbb{C}$ with the *bilinear* (not Hermitian) form $x_1y_1 + x_2y_2$, the line $\langle (1, i) \rangle \subset \mathbb{C}^2$ satisfies $W = W^\perp$. What is special about $\mathbb{F}_q$ is that Chevalley–Warning makes isotropy *unavoidable* and abundant.

5.2 Finite geometry

The totally isotropic subspaces of a nondegenerate form, ordered by inclusion, form a *polar space*: generalised quadrangles and their higher analogues, the geometries on which the classical groups act. Under positive-definiteness this geometry is empty.

5.3 Quantum error correction

Stabiliser codes are totally isotropic subspaces of $\mathbb{F}_2^{2n}$ under the standard symplectic form. The requirement that stabiliser generators commute is *exactly* the isotropy condition, and the quotient $C^\perp/C$ carries the logical operators. Symplectic $\mathbb{F}_2$-geometry is the native language of the Clifford group.

6 Where the analysis actually lives

If one wants Fourier analysis, the productive move is to stop trying to place an inner product on $\mathbb{F}_q^n$ and place it instead on the $\mathbb{C}$-vector space of functions:

$$\langle f, g \rangle \;=\; \sum_{x \in \mathbb{F}_q^n} f(x)\, \overline{g(x)}, \qquad f, g : \mathbb{F}_q^n \to \mathbb{C}.$$

This is a genuine positive-definite Hermitian inner product.

Definition 6.1 (Additive character)**.** An *additive character* of $\mathbb{F}_q$ is a group homomorphism $\psi : (\mathbb{F}_q, +) \to (\mathbb{C}^\times, \cdot)$. For $\mathbb{F}_p$ one may take $\psi(x) = e^{2\pi i x/p}$; in general $\psi(x) = e^{2\pi i \,\mathrm{Tr}(x)/p}$ with $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ the trace.

Fixing a nontrivial ψ, the functions $e_y(x) = \psi(\sum_i x_i y_i)$, for $y \in \mathbb{F}_q^n$, form an orthogonal basis, giving a Fourier transform, Plancherel's theorem and Poisson summation.

Observe the division of labour: the $\mathbb{F}_q$-bilinear form appears *inside the character*, indexing the basis, rather than defining the geometry of the space. The MacWilliams identity, relating the weight enumerator of a code C to that of $C^\perp$, is Poisson summation in exactly this setting — the bilinear form supplies the duality $C \leftrightarrow C^\perp$, and the $\mathbb{C}$-valued inner product supplies the analysis.

7 Summary of Part I

	$\mathbb{C}$	$\mathbb{R}$	$\mathbb{F}_q$ (q odd)
Nondegenerate symmetric forms	1 class	$n+1$ classes	2 classes
Nondegenerate alternating forms	1 class	1 class	1 class
Nondegenerate Hermitian forms	$n+1$ classes	—	1 class
$V \cong V^*$	yes	yes	yes
$V = W \perp W^\perp$	yes	yes	**no**
Anisotropic spaces	$\dim \le 1$	all n	$\dim \le 2$
Self-dual subspaces $W = W^\perp$	possible	no (definite)	**yes**
Norm, metric, analysis on V	yes	yes	no

Part II
Categories, lattices, and two kinds of opposite

Part I answered a question about forms. Part II asks a question of the same shape one level up, and the answer will turn out to be the same theorem.

The question. The opposite of the category of sets is the category of complete atomic Boolean algebras. What is the opposite of the category of vector spaces over $\mathbb{F}_q$?

8 Categories, from scratch

Definition 8.1 (Category)**.** A *category* $\mathcal{C}$ consists of:

- a collection $\mathrm{Ob}(\mathcal{C})$ of *objects*;
- for each pair X, Y of objects a set $\mathcal{C}(X, Y)$ of *morphisms* (written $f : X \to Y$);
- for each X an *identity* $\mathrm{id}_X \in \mathcal{C}(X, X)$;
- a *composition* $\mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \to \mathcal{C}(X, Z)$, $(g, f) \mapsto g \circ f$;

such that composition is associative and the identities are neutral: $h(gf) = (hg)f$, $f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f$.

Example 8.2. (a) **Set**: objects are sets, morphisms are functions. **FinSet**: the finite ones.
(b) $\mathbf{Vect}_k$: objects are k-vector spaces, morphisms are linear maps. $\mathbf{FinVect}_k$: the finite-dimensional ones.
(c) Any partially ordered set $(P, \leq)$ is a category: objects are the elements, and there is exactly one morphism $x \to y$ when $x \leq y$, none otherwise. Transitivity gives composition, reflexivity the identities.
(d) Any monoid is a category with one object, its elements being the morphisms.

Definition 8.3 (Isomorphism)**.** $f : X \to Y$ is an *isomorphism* if there is $g : Y \to X$ with $gf = \mathrm{id}_X$ and $fg = \mathrm{id}_Y$. In **Set** these are the bijections; in $\mathbf{Vect}_k$, the invertible linear maps.

Definition 8.4 (Opposite category)**.** The *opposite* $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$ and $\mathcal{C}^{\mathrm{op}}(X, Y) = \mathcal{C}(Y, X)$, with composition reversed. Formally: turn every arrow around. Note $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Taking the opposite is trivial *as an operation*; the content of a statement like "$\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$" is that the formally-reversed category, which has no obvious description, is equivalent to a category one already knows independently.

Definition 8.5 (Functor)**.** A *functor* $F : \mathcal{C} \to \mathcal{D}$ assigns an object FX to each object X and a morphism $Ff : FX \to FY$ to each $f : X \to Y$, preserving identities and composition: $F(\mathrm{id}_X) = \mathrm{id}_{FX}$, $F(g \circ f) = Fg \circ Ff$. A functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is called *contravariant* (as a rule on $\mathcal{C}$ it reverses arrows: $Ff : FY \to FX$, $F(gf) = Ff \circ Fg$).

Example 8.6 (The dual space functor)**.** $(-)^* : \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$ sends $V \mapsto V^*$ and $f : V \to W$ to the *transpose* $f^* : W^* \to V^*$, $\varphi \mapsto \varphi \circ f$. It reverses arrows because precomposition does.

Example 8.7 (The powerset functor)**.** $P : \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ sends $X \mapsto 2^X$ (the set of subsets) and $f : X \to Y$ to the *preimage* map $f^{-1} : 2^Y \to 2^X$. Again arrow-reversing, because preimage is a pullback operation. Note that the *image* map $f_* : 2^X \to 2^Y$ is also functorial

but *covariant*, and is not the right functor here: f^{-1} preserves all unions, intersections and complements, while f_* does not.

Definition 8.8 (Natural transformation)**.** Given functors $F, G : \mathcal{C} \to \mathcal{D}$, a *natural transformation* $\alpha : F \Rightarrow G$ is a family of morphisms $\alpha_X : FX \to GX$, one per object, such that for every $f : X \to Y$ the square

$$\begin{array}{ccc} FX & \xrightarrow{\alpha_X} & GX \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ FY & \xrightarrow[\alpha_Y]{} & GY \end{array}$$

commutes. It is a *natural isomorphism* if every α_X is an isomorphism.

This makes precise the remark in §1: for the double dual there is a natural transformation $\mathrm{Id} \Rightarrow (-)^{**}$, given by $v \mapsto \mathrm{ev}_v$ where $\mathrm{ev}_v(\varphi) = \varphi(v)$, and it is an isomorphism on finite-dimensional spaces. For the single dual no such transformation can even be written down: Id and $(-)^*$ have different variance, so the square above does not typecheck.

Definition 8.9 (Equivalence, anti-equivalence)**.** Functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ form an *equivalence* $\mathcal{C} \simeq \mathcal{D}$ if $GF \cong \mathrm{Id}_{\mathcal{C}}$ and $FG \cong \mathrm{Id}_{\mathcal{D}}$ naturally. An *anti-equivalence* or *duality* between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence $\mathcal{C}^{\mathrm{op}} \simeq \mathcal{D}$.

Equivalence, not isomorphism, is the correct notion of sameness for categories: it allows objects to be matched up only up to isomorphism, which is all one ever wants.

Proposition 8.10 (Finite-dimensional self-duality)**.** $\mathbf{FinVect}_k^{\mathrm{op}} \simeq \mathbf{FinVect}_k$*, via* $V \mapsto V^*$*, with the natural isomorphism* $V \cong V^{**}$ *witnessing the round trip.*

This is the cheap answer to our question: over any field, finite-dimensional linear algebra is its own opposite. Nothing new appears. That is suspicious, because in **Set** something new *does* appear, and we should like to know why.

9 Order, lattices and Boolean algebras, from scratch

Definition 9.1 (Poset)**.** A *partially ordered set* (poset) is a set P with a relation $\leq$ that is reflexive ($x \leq x$), antisymmetric ($x \leq y$ and $y \leq x$ imply $x = y$) and transitive.

Definition 9.2 (Meet and join)**.** In a poset, the *join* $x \vee y$ is the least upper bound of x and y (if it exists) and the *meet* $x \wedge y$ is the greatest lower bound. A *lattice* is a poset in which all binary meets and joins exist. It is *bounded* if it has a least element 0 and greatest element 1, and *complete* if *all* subsets (not just finite ones) have meets and joins.

Example 9.3. (a) 2^X, the subsets of a set X ordered by inclusion: $\vee = \cup$, $\wedge = \cap$, $0 = \emptyset$, $1 = X$. Complete.

(b) $L(V)$, the subspaces of a vector space ordered by inclusion: $W \wedge U = W \cap U$, $W \vee U = W + U$ (*not* $W \cup U$, which is not a subspace), $0 = \{0\}$, $1 = V$. Complete.

Definition 9.4 (Distributive, modular)**.** A lattice is *distributive* if

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z) \qquad \text{for all } x, y, z,$$

and *modular* if the weaker

$$x \leq z \implies x \vee (y \wedge z) = (x \vee y) \wedge z$$

holds. Distributive implies modular.

Proposition 9.5. *$L(V)$ is modular (Dedekind's modular law, valid for submodules of any module), and for $\dim V \geq 2$ it is not distributive.*

Proof of non-distributivity. Take three pairwise-distinct lines A, B, C in a plane. Then $B \vee C$ is the whole plane, so $A \wedge (B \vee C) = A$; but $A \wedge B = A \wedge C = 0$, so $(A \wedge B) \vee (A \wedge C) = 0 \neq A$. Three distinct lines exist in $\mathbb{F}_q^2$ for every $q \geq 2$, since a plane has $q + 1 \geq 3$ lines. □

This single computation is the seed of everything that follows: it is the exact point at which linear algebra departs from set theory, and it is visibly a statement about the presence of *more than two* one-dimensional subspaces — that is, about q.

Definition 9.6 (Complement, Boolean algebra)**.** In a bounded lattice, a *complement* of x is an element y with $x \wedge y = 0$ and $x \vee y = 1$. A lattice is *complemented* if every element has at least one complement. A *Boolean algebra* is a bounded distributive complemented lattice. In a Boolean algebra complements are unique, and one writes $\neg x$ or x^c.

Warning 9.7. *$L(V)$ is* complemented — every subspace has a direct complement — but the complement is far from unique, and $W \mapsto W^\perp$ is not in general one of them (that is exactly the isotropy problem of §3). Boolean-ness fails through *distributivity*, not through the existence of complements.

Definition 9.8 (Atom, atomic, CABA)**.** In a bounded lattice an *atom* is a minimal nonzero element. A complete lattice is *atomic* if every element is the join of the atoms below it. **CABA** is the category of *complete atomic Boolean algebras*, with morphisms the maps preserving arbitrary joins, arbitrary meets and complements.

Definition 9.9 (Boolean homomorphism, **Bool**, **FinBool**)**.** A *Boolean homomorphism* preserves $\vee, \wedge, \neg, 0, 1$. **Bool** is the category of Boolean algebras and homomorphisms; **FinBool** the finite ones.

Definition 9.10 (Filter, ultrafilter)**.** A *filter* on a Boolean algebra B is a nonempty upward-closed subset F closed under $\wedge$ with $0 \notin F$; it is an *ultrafilter* if for every x exactly one of $x, \neg x$ lies in F. Ultrafilters correspond exactly to homomorphisms $B \to 2$.

10 The duality $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$

Theorem 10.1 (Powerset duality)**.** *The functor $P : \mathbf{Set}^{\mathrm{op}} \to \mathbf{CABA}$, $X \mapsto 2^X$, $f \mapsto f^{-1}$, is an equivalence of categories. Its inverse sends a complete atomic Boolean algebra to its set of atoms. Restricting to finite objects,* $\mathbf{FinSet}^{\mathrm{op}} \simeq \mathbf{FinBool}$.

Sketch. 2^X is a complete atomic Boolean algebra whose atoms are the singletons, so $\mathrm{At}(2^X) \cong X$ naturally. Conversely, in a complete atomic Boolean algebra B every element is the join of the atoms beneath it, and this decomposition is unique, giving $B \cong 2^{\mathrm{At}(B)}$ naturally. Morphisms match because a complete Boolean homomorphism $2^Y \to 2^X$ is determined by where atoms go, i.e. by a function $X \to Y$. □

Remark 10.2 (Two neighbouring theorems). $\mathbf{Bool}^{\mathrm{op}} \simeq \mathbf{Stone}$, the category of *Stone spaces* (compact, Hausdorff, totally disconnected topological spaces), by sending B to its space of ultrafilters. This is Stone duality proper; Theorem 10.1 is the "complete/atomic" variant in which the topology becomes invisible. In the finite case all three coincide: $\mathbf{FinSet}^{\mathrm{op}} \simeq \mathbf{FinBool}$, and finite Stone spaces are just finite sets.

11 Why the two dualities differ: dualizing objects

Both $\mathbf{FinVect}^{\mathrm{op}} \simeq \mathbf{FinVect}$ and $\mathbf{FinSet}^{\mathrm{op}} \simeq \mathbf{FinBool}$ are instances of one pattern.

Definition 11.1 (Dualizing (schizophrenic) object)**.** Suppose an object D lives in two categories $\mathcal{C}$ and $\mathcal{D}$ "compatibly". Then $\mathrm{Hom}_{\mathcal{C}}(-, D)$ and $\mathrm{Hom}_{\mathcal{D}}(-, D)$ are contravariant functors that often form an adjunction, and restrict to a duality on the objects for which the canonical map $X \to \mathrm{Hom}(\mathrm{Hom}(X, D), D)$ is an isomorphism.

- In $\mathbf{Vect}_k$, the dualizing object is k: $V^* = \mathrm{Hom}_k(V, k)$. Here k is the *unit* of the tensor product and carries no internal structure beyond being the ground field, so $\mathrm{Hom}_k(V, k)$ is a bare vector space. The duality stays inside $\mathbf{Vect}_k$.
- In $\mathbf{Set}$, the dualizing object is $2 = \{\bot, \top\}$: $2^X = \mathrm{Hom}_{\mathbf{Set}}(X, 2)$. Here 2 is *not* the unit of the cartesian product (that is the one-point set 1); it is the *subobject classifier*, and it carries an internal Boolean algebra structure. So 2^X inherits that structure pointwise and the duality lands *outside* $\mathbf{Set}$, in $\mathbf{CABA}$.

Definition 11.2 (Subobject)**.** A morphism $m : U \to X$ is a *monomorphism* (mono) if $mf = mg \Rightarrow f = g$. A *subobject* of X is an isomorphism class of monos into X; $\mathrm{Sub}(X)$ denotes the poset of subobjects. In $\mathbf{Set}$, $\mathrm{Sub}(X) = 2^X$; in $\mathbf{Vect}_k$, $\mathrm{Sub}(V) = L(V)$.

Definition 11.3 (Subobject classifier, topos)**.** A *subobject classifier* in a category with finite limits is an object Ω with a mono $\mathsf{true} : 1 \to \Omega$ such that for every mono $U \rightarrowtail X$ there is a unique $\chi_U : X \to \Omega$ making

$$\begin{array}{ccc} U & \longrightarrow & 1 \\ \downarrowtail & & \downarrow{\scriptstyle \mathsf{true}} \\ X & \xrightarrow[\chi_U]{} & \Omega \end{array}$$

a pullback. An *(elementary) topos* is a category with finite limits, exponentials Y^X, and a subobject classifier. $\mathbf{Set}$ is a topos with $\Omega = 2$ and χ_U the indicator function.

So in $\mathbf{Set}$ the phrase "the dualizing object" and the phrase "the object that represents subobjects" name the *same* object 2. That coincidence is what makes $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$ simultaneously a hom-duality and a statement about subobject lattices. In $\mathbf{Vect}_k$ the coincidence fails — and it must.

Definition 11.4 (Additive and abelian categories)**.** A category is *additive* if each $\mathcal{C}(X, Y)$ is an abelian group with bilinear composition, there is a zero object 0 (both initial and terminal), and finite biproducts exist. It is *abelian* if additionally every morphism has a kernel and a cokernel, every mono is a kernel, and every epi is a cokernel. $\mathbf{Vect}_k$ is abelian.

Definition 11.5 (Strict initial object)**.** An initial object 0 is *strict* if every morphism $X \to 0$ is an isomorphism. In $\mathbf{Set}$ the initial object is $\emptyset$, and indeed any function $X \to \emptyset$ forces $X = \emptyset$. In every topos the initial object is strict.

Theorem 11.6. *No nontrivial category is both abelian and a topos.*

Proof. In an abelian category the initial and terminal objects coincide: $0 \cong 1$. In a topos the initial object is strict. So for any object X, the unique map $X \to 1 \cong 0$ is a morphism into a strict initial object, hence an isomorphism, hence $X \cong 0$ for every X and the category is equivalent to the one-object one-morphism category. □

The structural point. **Set** and $\mathbf{Vect}_q$ sit on opposite sides of a genuine dichotomy. Subobject lattices in a topos are Heyting algebras (Boolean, in **Set**); subobject lattices in an abelian category are modular. By Theorem 11.6 no category is both. Hence the **Set**-style duality and the **Vect**-style duality are not two instances of one theorem that one can interpolate naively: they are two different notions of "the dual", which happen to coincide only for **Set**.

12 Deforming the recipe: subspace lattices and projective geometry

The *content* of $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$ is: *a set is recovered from its lattice of subobjects, and that lattice is Boolean.* Let us run that recipe over $\mathbb{F}_q$.

Definition 12.1 (Projective space)**.** For a vector space V over k, the *projective space* $\mathbb{P}(V)$ has as points the 1-dimensional subspaces of V, as lines the 2-dimensional subspaces, and so on. When $\dim V = n$ and $k = \mathbb{F}_q$ one writes $\mathrm{PG}(n-1, q)$; its *projective dimension* is $n-1$.

Definition 12.2 (Gaussian binomial coefficient)**.** For $0 \le k \le n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \;=\; \prod_{i=0}^{k-1} \frac{q^{n-i}-1}{q^{k-i}-1}.$$

Proposition 12.3. *The number of k-dimensional subspaces of $\mathbb{F}_q^n$ is $\begin{bmatrix} n \\ k \end{bmatrix}_q$, and $\lim_{q\to 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}$.*

Example 12.4. $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = \frac{q^2-1}{q-1} = q+1$: a plane over $\mathbb{F}_q$ has $q+1$ lines, which is why Proposition 9.5 needed $q \ge 2$. At $q = 1$ this is $2 = \binom{2}{1}$.

Theorem 12.5 (Fundamental Theorem of Projective Geometry)**.** *Let V, W be vector spaces of dimension ≥ 3 over fields k, l. Every isomorphism of lattices $L(V) \to L(W)$ is induced by a σ-semilinear isomorphism $V \to W$ for some field isomorphism $\sigma : k \to l$, and the semilinear map is unique up to a scalar. In particular*

$$\mathrm{Aut}\big(L(V)\big) \;=\; \mathrm{P\Gamma L}(V) \;=\; \mathrm{PGL}(V) \rtimes \mathrm{Aut}(k).$$

This is the exact analogue of Theorem 10.1: *the object is recovered from its subobject lattice.* The difference is what kind of lattice one gets.

Category	$\mathrm{Sub}(X)$	Recovery theorem
Set	Boolean lattice 2^X	$\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$
$\mathbf{Vect}_{\mathbb{F}_q}$	projective geometry $L(V) = \mathrm{PG}(n-1, q)$	Fundamental Theorem (12.5)

Definition 12.6 (Quantum logic)**.** Birkhoff and von Neumann (1936) proposed reading the subspace lattice of a Hilbert space as a propositional logic: $\wedge$ = "and", $\vee$ = "or", $W^{\perp}$ = "not". Because the lattice is non-distributive, this *quantum logic* differs from classical propositional logic exactly in the distributive law. The failure computed in Proposition 9.5 is the same failure.

So: **the q-analogue of Bool is the category of projective geometries — equivalently, finite quantum logic.**

13 Dualities and polarities: where Part I reappears

We now identify the anti-automorphisms. In **Set** the anti-automorphism of 2^X is complementation $S \mapsto S^c$. What plays that role for $L(V)$?

Definition 13.1 (Correlation, duality, polarity)**.** An *anti-isomorphism* (or *duality*, or *correlation*) of lattices is an order-reversing bijection $\delta : L(V) \to L(W)$; equivalently an isomorphism $L(V)^{\mathrm{op}} \to L(W)$. A *polarity* is a duality $\delta : L(V) \to L(V)$ with $\delta^2 = \mathrm{id}$.

Definition 13.2 (Reflexive form)**.** A sesquilinear form (Definition 2.6) is *reflexive* if $\langle v, w\rangle = 0 \iff \langle w, v\rangle = 0$. Reflexivity is exactly what is needed for $W \mapsto W^\perp$ to be unambiguous (left and right perp agree).

Theorem 13.3 (Second Fundamental Theorem; Birkhoff–von Neumann, Baer)**.** *Let* $\dim V \geq 3$. *Every duality* $L(V) \to L(V)$ *is of the form* $W \mapsto W^\perp$ *for a nondegenerate sesquilinear form, unique up to a scalar. The dualities that are polarities correspond precisely to the reflexive forms, and every nondegenerate reflexive sesquilinear form is, after rescaling by a nonzero scalar, of exactly one of the following kinds:*

(i) symmetric bilinear ($\sigma = \mathrm{id}$, $\langle w, v\rangle = \langle v, w\rangle$)*;*
(ii) alternating bilinear ($\sigma = \mathrm{id}$, $\langle v, v\rangle = 0$)*;*
(iii) σ*-Hermitian* (σ *an involution* $\neq \mathrm{id}$)*.*

(In characteristic 2 *cases (i) and (ii) are not disjoint; see the remark in §2.)*

> **Part I and Part II are the same theorem.** The three families of forms classified in §2 — symmetric, alternating, Hermitian — are *not* a convenient list. By Theorem 13.3 they are exactly the implementations of "op" for linear algebra: the involutive anti-automorphisms of the subobject lattice. Asking "what replaces the inner product over $\mathbb{F}_q$?" and asking "what is the opposite of $\mathbf{Vect}_{\mathbb{F}_q}$?" are the same question, and the classification in Part I is the answer to both.

14 Complementation lost

Here the compressed sentence of §3 returns one level up.

Definition 14.1 (Ortholattice, orthomodular lattice)**.** An *orthocomplementation* on a bounded lattice is a map $x \mapsto x'$ with $x'' = x$, $x \leq y \Rightarrow y' \leq x'$, $x \wedge x' = 0$ and $x \vee x' = 1$. A lattice with one is an *ortholattice*; it is *orthomodular* if $x \leq y \Rightarrow y = x \vee (y \wedge x')$. Subspace lattices of Hilbert spaces are orthomodular; Boolean algebras are exactly the distributive ortholattices.

Now compare the three regimes for the map $W \mapsto W^\perp$:

	$\mathbb{C}$, positive definite	$\mathbb{F}_q$, nondegenerate	$\mathbb{F}_1$
order-reversing involution	yes	yes	yes
$\dim W + \dim W^\perp = \dim V$	yes	yes	yes
$W \wedge W^\perp = 0$	yes	**no**	yes
so: an orthocomplementation?	yes	**no**	yes
lattice distributive?	no	no	**yes**
resulting logic	orthomodular (quantum)	polarity only	Boolean (classical)

Over $\mathbb{F}_q$ one has *strictly less* than quantum logic: $W \mapsto W^\perp$ is an order-reversing involution but not an orthocomplementation, because $W \cap W^\perp \neq 0$ whenever W contains an isotropic vector. Boolean complementation is a polarity *plus* the complement property. Over $\mathbb{F}_q$ one gets the polarity and loses the complementation — which is precisely "one keeps the duality and loses the splitting", restated for lattices.

15 Why it is called quantum logic

The name is Birkhoff and von Neumann's, from *The Logic of Quantum Mechanics* (*Annals of Mathematics* **37**, 1936). "Logic" means the algebra of *propositions* about a physical system; "quantum" means the propositions that quantum mechanics permits one to assert. The claim being made is that physics determines which propositional calculus is correct.

15.1 Classical propositions form a Boolean algebra

Definition 15.1 (Phase space, classical proposition)**.** A classical mechanical system has a *phase space* X, the set of its possible states (positions and momenta). A *proposition* about the system is a yes/no question — "the particle lies in region R", "the energy is below E" — and is identified with the subset of X on which it holds.

Under that identification, "P and Q" is $P \cap Q$, "P or Q" is $P \cup Q$, and "not P" is P^c. So the propositions of a classical system form a Boolean algebra. Note that classical logic is not *assumed* here: it is *derived* from the facts that states are points and propositions are subsets.

15.2 Quantum propositions are subspaces

Definition 15.2 (Hilbert space)**.** A *Hilbert space* is a complex vector space H with a positive-definite Hermitian inner product that is complete in the induced norm. A subspace is *closed* if it contains all its limits; in finite dimensions every subspace is closed.

Definition 15.3 (Projection, self-adjoint operator)**.** The *adjoint* A^* of a bounded operator is defined by $\langle Av, w\rangle = \langle v, A^*w\rangle$; A is *self-adjoint* if $A = A^*$. A *(orthogonal) projection* is a self-adjoint P with $P^2 = P$; projections correspond bijectively to closed subspaces, $P \leftrightarrow \operatorname{im} P$.

Theorem 15.4 (Spectral theorem, projection-valued form)**.** *Every self-adjoint operator A determines a projection-valued measure E_A: to each Borel set $\Delta \subseteq \mathbb{R}$ it assigns a projection $E_A(\Delta)$, additively on disjoint unions, with $E_A(\mathbb{R}) = \mathrm{id}$.*

The projection $E_A(\Delta)$ *is* the proposition "the value of the observable A lies in Δ". Hence:

> Quantum propositions are the closed subspaces of H, and
>
> $$\text{“and”} = W \cap U, \qquad \text{“or”} = \overline{W + U}, \qquad \text{“not”} = W^\perp.$$

A state ψ makes the proposition W *definitely true* if $\psi \in W$, *definitely false* if $\psi \in W^\perp$, and otherwise assigns it only the probability $\|P_W \psi\|^2$ (the *Born rule*).

The decisive entry is the middle one. "Or" is *not* union, because a union of subspaces is not a subspace; one must take the closed span, and the span contains vectors lying in neither disjunct. Those are the superpositions.

15.3 The failure of distributivity, read physically

Take a qubit, $H = \mathbb{C}^2$, with the usual notation $|0\rangle, |1\rangle$ for the z-spin eigenstates and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ for an x-spin eigenstate. Put

$$a = \text{“spin is } +x\text{”} = \langle |+\rangle \rangle, \quad b = \text{“spin is } +z\text{”} = \langle |0\rangle \rangle, \quad c = \text{“spin is } -z\text{”} = \langle |1\rangle \rangle.$$

Then $b \vee c = H$, the tautology “the z-spin is up or down”, so

$$a \wedge (b \vee c) = a \wedge 1 = a, \qquad \text{but} \qquad (a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0.$$

In words: “the spin is $+x$, *and* the z-spin is either up or down” is true, since the second conjunct is a tautology; whereas “either (the spin is $+x$ and z-up) or (the spin is $+x$ and z-down)” is a disjunction of two contradictions, there being no simultaneous eigenstate of x- and z-spin. Distributing across the disjunction converts a truth into a falsehood.

The physical content is: *a disjunction can be true without either disjunct being true.* $|+\rangle$ lies in the span of $\{|0\rangle, |1\rangle\}$ but in neither. Superposition, stated as a fact about a lattice.

Remark 15.5 (This is Proposition 9.5). The computation just performed is the one proving $L(V)$ non-distributive: three pairwise-distinct lines A, B, C in a plane satisfy $A \wedge (B \vee C) = A$ while $(A \wedge B) \vee (A \wedge C) = 0$. Here the plane is $\mathbb{C}^2$ and the three lines are $|0\rangle$, $|1\rangle$, $|+\rangle$.

Over $\mathbb{F}_2$ a plane has exactly three lines, spanned by $(1,0)$, $(0,1)$ and $(1,1)$ — that is, by $|0\rangle$, $|1\rangle$ and $|0\rangle + |1\rangle$. So the smallest non-distributive quantum logic is the $\mathbb{F}_2$ qubit, and the hypothesis “$q \geq 2$” in the proof of Proposition 9.5 is exactly the hypothesis that superposition exists at all.

15.4 Is it really *logic*? The case against

As a logic it is impoverished, and this should be said plainly.

- There is no satisfactory conditional. The deduction theorem fails, and the candidate implications (the Sasaki hook $a \to b := a' \vee (a \wedge b)$ and its rivals) each fail some property one wants.
- Orthomodular lattices have no well-behaved tensor product, so composing two systems — the very place quantum mechanics is most distinctive — is awkward.
- A deflationary reading is available: nothing about *inference* has changed; what changed is which propositions are simultaneously meaningful. On that reading the subject is projective geometry with a suggestive vocabulary.

Putnam’s *Is Logic Empirical?* (1968) argued for the strong reading, that quantum mechanics revises logic as relativity revised geometry. He later retracted it.

15.5 The case for: the lattice determines the physics

The name is nonetheless earned, because the propositional lattice turns out to contain the entire theory — a startling amount of content for a partially ordered set.

Theorem 15.6 (Gleason, 1957)**.** *Let* $\dim H \geq 3$. *Every countably additive probability measure on the lattice of projections of* H *has the form* $P \mapsto \operatorname{tr}(\rho P)$ *for a unique density operator* ρ.

So the Born rule need not be postulated: it is forced by the lattice together with countable additivity. (The hypothesis $\dim H \geq 3$ is essential — the qubit used above admits exotic measures — and it is the same dimension bound as in the Fundamental Theorem 12.5.)

Definition 15.7 (Division ring, quaternions)**.** A *division ring* satisfies all the field axioms except commutativity of multiplication. The *quaternions* $\mathbb{H}$ form the four-dimensional real division ring with basis $1, i, j, k$ and $i^2 = j^2 = k^2 = ijk = -1$, with conjugation $\overline{a + bi + cj + dk} = a - bi - cj - dk$ as involution.

Theorem 15.8 (Piron 1964; Solèr 1995)**.** *A complete, atomistic, irreducible orthomodular lattice with the covering property and dimension ≥ 4 is isomorphic to the lattice of closed subspaces of a Hermitian space over some division ring with involution. If it moreover admits an infinite orthonormal sequence, that division ring is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ with its standard involution, and the space is a Hilbert space.*

Piron's step is, in essence, the Fundamental Theorem of Projective Geometry (Theorem 12.5) together with the classification of polarities (Theorem 13.3). Which means the three families of Part I reappear a third time — now as the complete list of candidate quantum mechanics:

polarity type (Thm. 13.3)	division ring, involution	physics
symmetric bilinear	$\mathbb{R}$, trivial involution	real quantum mechanics
Hermitian	$\mathbb{C}$, complex conjugation	standard quantum mechanics
Hermitian	$\mathbb{H}$, quaternionic conjugation	quaternionic quantum mechanics
alternating	—	**excluded**

Proposition 15.9 (Why symplectic quantum mechanics is impossible)**.** *An alternating form satisfies $\langle v, v\rangle = 0$ for every v, so every subspace meets its own perp: $W \wedge W^{\perp} \neq 0$ for all $W \neq 0$. Hence $W \mapsto W^{\perp}$ is never an orthocomplementation (Definition 14.1), and the lattice is never an ortholattice.*

Positive-definiteness is what selects two of the three families and discards the symplectic one. The classification of §2 is simultaneously: (i) the answer to "what replaces the inner product over $\mathbb{F}_q$", (ii) by Theorem 13.3, the list of implementations of "op" for linear algebra, and (iii) by Theorem 15.8, the list of possible quantum mechanics. One classification, three readings.

15.6 Over $\mathbb{F}_q$: the logic without the statistics

What does the finite-field case inherit? Exactly the lattice, and nothing measure-theoretic. The obstruction is Remark 1.7: no ordering, hence no probabilities.

kept over $\mathbb{F}_q$	lost over $\mathbb{F}_q$
superposition (Remark 15.5)	the Born rule: $\|P_W\psi\|^2$ needs $[0,1] \subset \mathbb{R}$
non-distributivity	Gleason's theorem: no measures to classify
entanglement, no-cloning, contextuality	orthomodularity itself (table in §14)

Amplitudes exist over $\mathbb{F}_q$; magnitudes do not. This is a live research area — *modal quantum theory* (Schumacher and Westmoreland), quantum mechanics over finite fields, in which the only predicates are *possible* and *impossible* rather than probabilities. Remarkably, analogues of teleportation, superdense coding, no-cloning and a Bell-type contextuality theorem all survive there. It is a clean experiment separating the structural features of quantum mechanics from those requiring probability.

15.7 A wrinkle: von Neumann's own retreat

Von Neumann grew dissatisfied with the projection lattice of $B(H)$. For infinite-dimensional H that lattice is orthomodular but *not modular* (Definition 9.4), and modularity is what supports a well-behaved dimension function. Wanting a dimension taking values continuously

in $[0,1]$ — a canonical *a priori* probability — he was led to type II_1 factors and continuous geometry, and away from quantum logic toward what are now called von Neumann algebras.

The wrinkle for us: over $\mathbb{F}_q$ every space is finite-dimensional, so $L(V)$ *is* modular (Proposition 9.5). The finite-field case sits on von Neumann's preferred side of that split. It has the modularity he wanted, and lacks — for the entirely separate reason that $\mathbb{F}_q$ is unordered — the measure he wanted it for.

Part III
The limit $q \to 1$

16 The field with one element

Warning 16.1. There is no field with one element: a field requires $1 \neq 0$. "$\mathbb{F}_1$" names a programme, not an object — the search for a base beneath $\operatorname{Spec}\mathbb{Z}$ over which q-analogue formulas make sense at $q = 1$. Implementations include Tits' original observation (1957), Deitmar's monoid schemes, Kapranov–Smirnov, Connes–Consani, Borger's Λ-rings and Toën–Vaquié. They agree on the combinatorics below and differ elsewhere.

Definition 16.2 (Pointed set; the standard model)**.** A *pointed set* is a pair $(X, *)$ with $* \in X$; a morphism of pointed sets preserves the basepoint. $\mathbf{FinSet}_*$ denotes the category of finite pointed sets. In the Deitmar / Kapranov–Smirnov model, an $\mathbb{F}_1$*-vector space* is a finite pointed set (the basepoint is 0), and an $\mathbb{F}_1$*-linear map* is a pointed map. There is no addition — $\mathbb{F}_1$ has no additive structure to inherit — and scalar multiplication is trivial because $\mathbb{F}_1^\times = \{1\}$.

Proposition 16.3 (GL at $q = 1$)**.** $\mathrm{GL}_n(\mathbb{F}_1) = S_n$*, the symmetric group: the* $\mathbb{F}_1$*-linear automorphisms of a pointed set with* n *non-basepoint elements are the permutations of those elements. Numerically,*

$$|\,\mathrm{GL}_n(\mathbb{F}_q)| \;=\; \prod_{i=0}^{n-1}\bigl(q^n - q^i\bigr) \;=\; q^{\binom{n}{2}}(q-1)^n\,[n]_q!, \qquad [n]_q! = \prod_{i=1}^{n}\frac{q^i-1}{q-1},$$

so after dividing out the n *copies of the scalar group* $\mathbb{F}_q^\times$ *— which degenerates to the trivial group — and setting* $q = 1$ *one gets* $n!$*.*

Remark 16.4 (Tits' observation)*.* The building of $\mathrm{GL}_n(\mathbb{F}_q)$ is the flag complex of the subspace lattice $L(\mathbb{F}_q^n)$; at $q = 1$ it degenerates to the Coxeter complex of S_n, i.e. to a single apartment. "The building becomes the apartment" and "the subspace lattice becomes the Boolean lattice" are the same statement: a frame (unordered basis up to scalars) is exactly a set.

17 Isotropy vanishes, and Boolean logic reappears

Proposition 17.1 (Counts of nonzero isotropic vectors)**.** *For a nondegenerate form on a space over* $\mathbb{F}_q$*, the number of nonzero isotropic vectors is:*

form	*count*	*value at* $q = 1$
symplectic $\mathrm{Sp}_{2m}(q)$	$q^{2m} - 1$	0
orthogonal $O^+_{2m}(q)$	$(q^m - 1)(q^{m-1} + 1)$	0
orthogonal $O^-_{2m}(q)$	$(q^m + 1)(q^{m-1} - 1)$	0
orthogonal $O_{2m+1}(q)$	$q^{2m} - 1$	0
unitary $U_n(q)$	$\bigl(q^n - (-1)^n\bigr)\bigl(q^{n-1} - (-1)^{n-1}\bigr)$	0

Every one of them vanishes at $q = 1$*.*

Why this must happen. Isotropy requires *addition*. To have $\langle v, v\rangle = 0$ with $v \neq 0$ one needs a sum of nonzero terms to cancel, and $\mathbb{F}_1$ has no addition at all. The vanishing of the counts is the arithmetic shadow of that structural absence. □

Corollary 17.2 (The chain closes)**.** *At* $q = 1$*:*

(a) *there are no isotropic vectors, so* $W \wedge W^{\perp} = 0$ *and the polarity* $W \mapsto W^{\perp}$ *becomes a genuine complementation;*
(b) *the subspace lattice* $L_n(q)$ *becomes the Boolean lattice* $2^{[n]}$*, hence distributive;*
(c) *distributive + complemented = Boolean.*

Therefore the q-analogue of **Bool** *degenerates to* **Bool**, *and* $\mathbf{Set}^{\mathrm{op}} \simeq \mathbf{CABA}$ *is recovered:*

$$\mathbf{Vect}_{\mathbb{F}_q} \xrightarrow{\text{subobject lattices}} \text{projective geometry / quantum logic} \xrightarrow{q \to 1} \mathbf{Bool}.$$

The $q \to 1$ dictionary

	at q	at $q = 1$
object	$\mathbb{F}_q$-vector space $\mathbb{F}_q^n$	pointed set, n non-base points
number of k-subobjects	$\begin{bmatrix} n \\ k \end{bmatrix}_q$	$\binom{n}{k}$
subobject lattice	$L_n(q)$: modular, not distributive	$2^{[n]}$: Boolean
points of $\mathrm{PG}(n-1, q)$	$1 + q + \cdots + q^{n-1}$	n
automorphisms of the lattice	$\mathrm{P\Gamma L}_n(q)$	S_n
nonzero isotropic vectors	see Prop. 17.1	0
$W \mapsto W^{\perp}$	polarity, not a complementation	Boolean complementation
building	Tits building of $\mathrm{GL}_n(\mathbb{F}_q)$	Coxeter complex of S_n

18 An aside: the prime at infinity is not $\mathbb{F}_1$

Definition 18.1 (Absolute value, place)**.** An *absolute value* on a field k is $|\cdot| : k \to \mathbb{R}_{\geq 0}$ with $|x| = 0 \iff x = 0$, $|xy| = |x||y|$ and $|x + y| \leq |x| + |y|$. Two are *equivalent* if one is a positive power of the other; an equivalence class is a *place*. The *p-adic* absolute value is $|p^a u/v|_p = p^{-a}$ for u, v coprime to p.

Theorem 18.2 (Ostrowski)**.** *Every nontrivial place of* $\mathbb{Q}$ *is either* $|\cdot|_p$ *for a prime* p*, or the usual archimedean* $|\cdot|_\infty$*. The completions are* $\mathbb{Q}_p$ *and* $\mathbb{R}$*, and the product formula* $\prod_v |x|_v = 1$ *holds for* $x \in \mathbb{Q}^\times$*.*

The place ∞ is thus a genuine place of $\mathbb{Q}$, "the prime at infinity", with completion $\mathbb{R}$ and residue characteristic 0. It is an *archimedean completion*. $\mathbb{F}_1$, by contrast, is a proposed *degeneration*: a base such that $\operatorname{Spec}\mathbb{Z} \to \operatorname{Spec}\mathbb{F}_1$ is terminal, so that the fibre product $\operatorname{Spec}\mathbb{Z} \times_{\operatorname{Spec}\mathbb{F}_1} \operatorname{Spec}\mathbb{Z}$ exists and a Weil-style proof of the Riemann Hypothesis might be imitated (Deninger, Manin). These are different directions: one completes, the other degenerates.

They are nonetheless productively confused, because both are "the extra point one bolts onto $\operatorname{Spec}\mathbb{Z}$", and because Connes–Consani genuinely knit them together via the adèle class space, where the archimedean place and the $\mathbb{F}_1$-structure appear in one construction. The confusion is fertile; it is still a confusion.

19 Three answers, preserved

The phrase "the opposite of **Set** is **Bool**" bundles two claims which coincide only because **Set** is a topos. Unbundled, the q-deformation goes three ways, and I do not think they reconcile.

(1) The literal answer: linear algebra is self-dual.
$\mathbf{FinVect}_q^{\mathrm{op}} \simeq \mathbf{FinVect}_q$ (Proposition 8.10). For infinite dimensions one needs topology:

Definition 19.1 (Linearly compact vector space)**.** A topological k-vector space (k discrete) is *linearly topologised* if 0 has a neighbourhood basis of open subspaces, and *linearly*

compact if in addition it is Hausdorff, complete, and every family of closed cosets with the finite-intersection property has nonempty intersection. Equivalently: it is an inverse limit of finite-dimensional discrete spaces.

Theorem 19.2 (Lefschetz duality)**.** $\mathbf{Vect}_k^{\mathrm{op}} \simeq \mathbf{LinComp}_k$*, via* $V \mapsto V^*$ *with the topology of pointwise convergence.*

Definition 19.3 (Ind and Pro)**.** $\mathrm{Ind}(\mathcal{C})$ is the category of formal filtered colimits of objects of $\mathcal{C}$; $\mathrm{Pro}(\mathcal{C}) = \mathrm{Ind}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}$, formal cofiltered limits.

Since $\mathbf{Vect}_k = \mathrm{Ind}(\mathbf{FinVect}_k)$ we get $\mathbf{Vect}_k^{\mathrm{op}} = \mathrm{Pro}(\mathbf{FinVect}_k^{\mathrm{op}}) = \mathrm{Pro}(\mathbf{FinVect}_k) = \mathbf{LinComp}_k$; and since $\mathbf{Set} = \mathrm{Ind}(\mathbf{FinSet})$ we get $\mathbf{Set}^{\mathrm{op}} = \mathrm{Pro}(\mathbf{FinBool}) \simeq \mathbf{CABA}$, while $\mathbf{Bool}^{\mathrm{op}} = \mathrm{Pro}(\mathbf{FinSet}) = \mathbf{Stone}$. All three dualities are the same formal move — pass to the pro-completion — and in that sense $\mathbf{LinComp}_{\mathbb{F}_q}$ *is* the q-analogue of **CABA**.

(2) The structural answer: projective geometry.
Deform the recipe rather than the statement. Subobject lattices give $\mathbf{Set} \rightsquigarrow \mathbf{Bool}$ and $\mathbf{Vect}_q \rightsquigarrow \mathrm{PG}(n-1, q)$; recovery is Theorem 12.5 in place of Theorem 10.1; the anti-automorphisms are polarities, classified by Theorem 13.3 into exactly the three families of Part I; and the whole thing degenerates to **Bool** at $q = 1$. This is the answer that makes the question's chain close.

(3) The answer at $q = 1$ itself.
The opposite of $\mathbb{F}_1$-linear algebra is $\mathbf{FinSet}_*^{\mathrm{op}}$, which is Segal's category Γ — the indexing category for Γ-spaces and hence for E_∞ (coherently commutative) structures in homotopy theory. Concretely:

Proposition 19.4. $\mathbf{FinSet}_*^{\mathrm{op}} \simeq \mathbf{FinBool}_{/2}$*, the category of finite Boolean algebras equipped with a homomorphism to* 2*, i.e. with a distinguished ultrafilter.*

Proof. $\mathbf{FinSet}_* = \mathbf{FinSet}_{1/}$, the coslice under the one-point set. Opposites exchange coslice and slice: $(\mathcal{C}_{1/})^{\mathrm{op}} = (\mathcal{C}^{\mathrm{op}})_{/1}$. Under $\mathbf{FinSet}^{\mathrm{op}} \simeq \mathbf{FinBool}$ the object 1 corresponds to $P(1) = 2$. Finally a homomorphism $B \to 2$ is the same as an ultrafilter on B. □

So even the basepoint — the 0 of $\mathbb{F}_1$-linear algebra — has a Boolean shadow.

Mai nafka minah. Answer (1) is right if one asks *what has the same shape*; answer (2) is right if one asks *what does the same job*; answer (3) is right if one asks *what happens at the degenerate point*. They diverge because **Set** conflates its dualizing object with its subobject classifier and $\mathbf{Vect}_q$ cannot, by Theorem 11.6. The disagreement should be kept, not resolved.

Coda

q general	$\longrightarrow$	$q = 1$
$\mathbf{Vect}_{\mathbb{F}_q}$		$\mathbf{FinSet}_*$
projective geometry $\mathrm{PG}(n-1, q)$		Boolean algebra $2^{[n]}$
polarity $W \mapsto W^\perp$		complementation $S \mapsto S^c$
symmetric / alternating / Hermitian		(isotropy empty)
quantum logic		classical logic

One keeps the duality and loses the splitting; at $q = 1$ the splitting comes back, and with it, Boole.